

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 5: Total Drag, Minimum Drag Speed (V_{md}) & Ground Effect

Syllabus Source: CAE Oxford (pp. 145-184) | Pass Target: $\geq 90\%$ AviationExam

1. PARASITE DRAG COMPONENTS: SKIN FRICTION, FORM & INTERFERENCE

Parasite drag includes all aerodynamic drag not associated with the generation of lift. It increases with the square of airspeed (D_p proportional to V^2):

- **Skin Friction Drag:** Result of viscous shearing stresses in the boundary layer. Minimized by keeping surfaces polished and maintaining laminar flow.
- **Form Drag (Pressure Drag):** Result of boundary layer separation creating a low-pressure wake behind bluff bodies. Minimized by streamlining and optimizing fineness ratio (length-to-diameter ratio $\sim 3:1$ to $4:1$).
- **Interference Drag:** Result of turbulent mixing of boundary layers at structural junctions (wing root, fuselage, engine pylons). Minimized by aerodynamic fairings and root fillets.

2. TOTAL DRAG CURVE & MINIMUM DRAG SPEED (VMD)

Total drag is the sum of Induced Drag (D_i proportional to $1/V^2$) and Parasite Drag (D_p proportional to V^2):

- **Vmd Equality Point:** At Minimum Drag Speed (V_{md}), INDUCED DRAG EXACTLY EQUALS PARASITE DRAG: $D_i = D_p = 1/2$ Total Drag!
- **Speed Stability Regimes:** Speeds above V_{md} : Speed-stable (parasite drag dominates; slowing down reduces drag). Speeds below V_{md} : The 'Backside of the Drag Curve' / Reverse Command (induced drag dominates; flying slower REQUIRES MORE THRUST to maintain level flight!).
- **Weight Variation:** Increasing aircraft gross mass (W) shifts the entire total drag curve UP and to the RIGHT: V_{md} increases with the square root of mass (V_{md} proportional to $\sqrt{W}$); Minimum Total Drag increases directly proportional to mass (Total Drag proportional to W).

3. GROUND EFFECT AERODYNAMICS

When an aircraft flies within one wingspan distance above the ground ($h \leq b$, especially pronounced at $h \leq b/4$):

Aerodynamic Parameter	Physical Change Entering Ground Effect	Cockpit Symptoms & Pilot Action
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Induced Downwash & Vortices	Ground physically blocks vertical flow, suppressing wingtip vortices and reducing downwash angle.	Induced drag DROPS by up to 50%!
Effective Angle of Attack	Reduction of downwash rotates lift vector forward toward the vertical.	Lift INCREASES significantly at the same geometric pitch attitude.
Elevator Control Effectiveness	Reduced downwash over the tailplane decreases horizontal stabilizer download.	Generates an uncommanded NOSE-DOWN pitching moment on landing!
Airspeed Indicator Static Vent	Local static pressure at the vent increases slightly.	ASI and Altimeter read slightly LOWER than actual in ground effect.

[!] EXAM TRAP (AVIATIONEXAM) - Ground Effect Traps on Take-off and Landing

- LANDING (Floating): Due to the sudden reduction in induced drag, an aircraft entering flare with excess airspeed will FLOAT down the runway, risking runway overrun! - TAKE-OFF (Premature Liftoff Stall): An aircraft can lift off prematurely at a dangerously low speed inside ground effect. As it climbs out of ground effect (above $h > b/2$), induced drag suddenly increases and effective AOA drops, causing the aircraft to stall or settle back onto the runway!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Speed Instability & Ground Effect**

Rule 1 (Back Side of Drag Curve): If flying below V_{MD} on approach, do NOT pull back on the yoke to recover altitude if speed is decaying. Below V_{MD} , slower speed increases total drag. You must add thrust immediately. Rule 2 (Premature Liftoff Trap): In ground effect, induced drag drops by up to 50%. The aircraft may lift off below normal rotation speed V_r . Upon climbing out of ground effect (above half wingspan), induced drag surges, risking an immediate stall.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: At the Minimum Drag Speed (V_{md}) in straight and level flight, what is the exact mathematical relationship?

- [A]: Induced Drag is twice Parasite Drag
- [B]: Induced Drag EXACTLY EQUALS Parasite Drag ($D_i = D_p = 50\%$ of Total Drag)
- [C]: Induced Drag is zero
- [D]: Parasite Drag is zero

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. At the minimum point of the total drag curve (V_{md}), the decreasing induced drag curve intersects the increasing parasite drag curve. Induced drag and parasite drag are exactly equal.

Q2: How does an increase in aircraft gross mass affect the Minimum Drag Speed (V_{md}) and minimum total drag?

- [A]: V_{md} decreases and drag increases
- [B]: Both V_{md} and minimum total drag INCREASE
- [C]: V_{md} increases but drag remains constant
- [D]: Neither changes

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Increasing weight requires higher lift. Higher lift requires higher speed to minimize drag (V_{md} increases proportional to $\sqrt{W}$) and produces higher total drag at that speed.

Q3: What occurs aerodynamically when an aircraft enters Ground Effect during landing flare?

- [A]: Induced drag increases and lift decreases
- [B]: Induced drag drops significantly, effective AOA increases, and the aircraft experiences an uncommanded nose-down pitching moment
- [C]: Skin friction drag triples

- [D]: The stall speed increases

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The proximity of the runway suppresses wingtip vortices and downwash, slashing induced drag, boosting lift (causing floating), and reducing tailplane downforce (causing a nose-down pitch).